

Order Flow Imbalance and the Decay of Price Impact in CME Ether Futures

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Abstract

We study Order Flow Imbalance (Cont, Kukanov, and Stoikov, 2014) on CME Ether futures over February 2021 to August 2026.

On 28,711 out-of-sample dollar-volume bars, an OLS regression of bar mid-quote change on contemporaneous OFI with Newey-West HAC standard errors yields $\hat{\beta} = +0.215$ ($t = 58.1$, $R^2 = 0.30$).

Measured directly from the raw tick stream, the signal’s forward impact peaks near +11.3 bp at four hours; it is significant at every horizon out to that peak under a by-day block bootstrap and indistinguishable from zero beyond it, and a sign-randomised placebo stays within ± 1.3 bp of zero throughout.

That peak is comparable to the cost of capturing it, since the contract’s quoted spread runs 10 to 30 ticks, so OFI is informative about price on this contract without being straightforwardly tradable.

1 Introduction

Order Flow Imbalance (OFI) aggregates best-bid and best-offer quote arrivals into a signed measure of top-of-book pressure. On liquid US equities, contemporaneous OFI explains roughly two-thirds of bar-level mid-quote variation (Cont, Kukanov, and Stoikov, 2014). The cryptocurrency evidence is thinner: Silantsev (2019) finds trade-flow imbalance dominates short-horizon prices on BitMEX XBT/USD, and Bieganski and Ślepaczuk (2026) report OFI as the top SHAP feature in CatBoost ensembles on Binance Futures perpetuums. CME’s Ether contract sits between these regimes: a regulated venue with thinner top-of-book depth than the equity tape and a higher tick-relative spread than either.

This paper estimates the contemporaneous OFI signal-impact on CME ETH futures from February 2021 to August 2026 and characterises how that impact decays. The construction is the level-1 (best bid / best offer) form of Cont, Kukanov, and Stoikov on dollar-volume bars built from TBBO (trade-sampled top-of-book) events (Lopez de Prado, 2018). Forward impact is then measured directly at sub-bar to 24-hour horizons.

2 Data

The panel consists of CME-cleared TBBO data for the front-month ETH futures contract, covering February 2021 through August 2026, with rolls handled by contract stitching.

Bars are constructed in dollar volume, with the threshold calibrated so that a median in-sample trading day produces about ten bars, and that threshold is held fixed when bars are constructed for the out-of-sample period.

*Independent research conducted by the author; the views expressed are the author’s own and not those of Imperial College London.

For each bar the following quantities are computed:

- **ofi**: signed level-1 (best bid / best offer) order flow imbalance, following the construction of Cont, Kukanov, and Stoikov (2014). For each TBBO update i , let p_i^B, p_i^A denote the best-bid and best-ask quotes and q_i^B, q_i^A their sizes. The per-event OFI contribution is $e_i = e_i^B + e_i^A$, where

$$e_i^B = \begin{cases} +q_i^B & \text{if } p_i^B > p_{i-1}^B \\ -q_{i-1}^B & \text{if } p_i^B < p_{i-1}^B \\ q_i^B - q_{i-1}^B & \text{if } p_i^B = p_{i-1}^B \end{cases}, \quad e_i^A = \begin{cases} -q_i^A & \text{if } p_i^A < p_{i-1}^A \\ +q_{i-1}^A & \text{if } p_i^A > p_{i-1}^A \\ -(q_i^A - q_{i-1}^A) & \text{if } p_i^A = p_{i-1}^A. \end{cases}$$

The bar OFI is the unnormalised sum over all events in the bar, $O_t = \sum_{i \in \text{bar } t} e_i$.

- **mid_close**: the mid quote $(p^B + p^A)/2$ at the bar boundary.
- **d_mid**: the change in mid quote since the previous bar.

3 Signal-impact regression

Following Cont, Kukanov, and Stoikov (2014) and Silantsev (2019), bar mid-quote change is regressed on contemporaneous and one-bar lagged OFI:

$$\Delta p_t = \alpha + \beta_0 O_t + \beta_1 O_{t-1} + \varepsilon_t, \quad (1)$$

where $\Delta p_t = m_t - m_{t-1}$ is the change in mid-quote between bar boundaries and O_t is the bar OFI. Standard errors are Newey-West HAC (Newey and West, 1987) with five lags. Table 1 reports the estimates on the full panel, the 2021-02 to 2023-12 in-sample (IS) window, and the 2024-01 to 2026-08 out-of-sample (OOS) window.

Window	n (bars)	$\hat{\beta}_0$	HAC $t(\beta_0)$	$\hat{\beta}_1$	HAC $t(\beta_1)$	R^2
Full sample	41,055	+0.247	+70.2	-0.027	-14.4	0.29
In-sample 2021-23	12,344	+0.300	+44.9	-0.025	-7.0	0.31
Out-of-sample 2024-26	28,711	+0.215	+58.1	-0.026	-13.3	0.30

Table 1: OLS, Newey-West HAC (five lags). $p < 10^{-6}$ on both coefficients in every window.

Both subsamples reject $\beta_0 = 0$ at $p < 10^{-6}$. The IS R^2 of 0.31 and OOS R^2 of 0.30 match to within rounding, and the slope coefficient moves from +0.300 in-sample to +0.215 out-of-sample. Both R^2 figures are below the two-thirds level of Cont et al. (2014) for liquid US equities; CME Ether futures have lower depth and a higher tick-relative spread.

The one-bar-lagged term is negative, at roughly 12% of the contemporaneous impact (Figure 1). The regression is identified at bar resolution (mean OOS bar duration ≈ 47 minutes); Section 3.1 characterises the signal's forward impact at sub-bar to 24-hour horizons directly.

3.1 Signal decay profile

The bar regression cannot resolve signal decay at sub-bar horizons. Forward impact is measured directly at 21 horizons from 30 seconds to 24 hours. A signal event is a bar at which the rolling z -score of cumulative OFI over $K = 5$ bars,

$$z_t = \frac{C_t - \mu_t}{\sigma_t}, \quad C_t = \sum_{s=t-K+1}^t O_s, \quad (2)$$

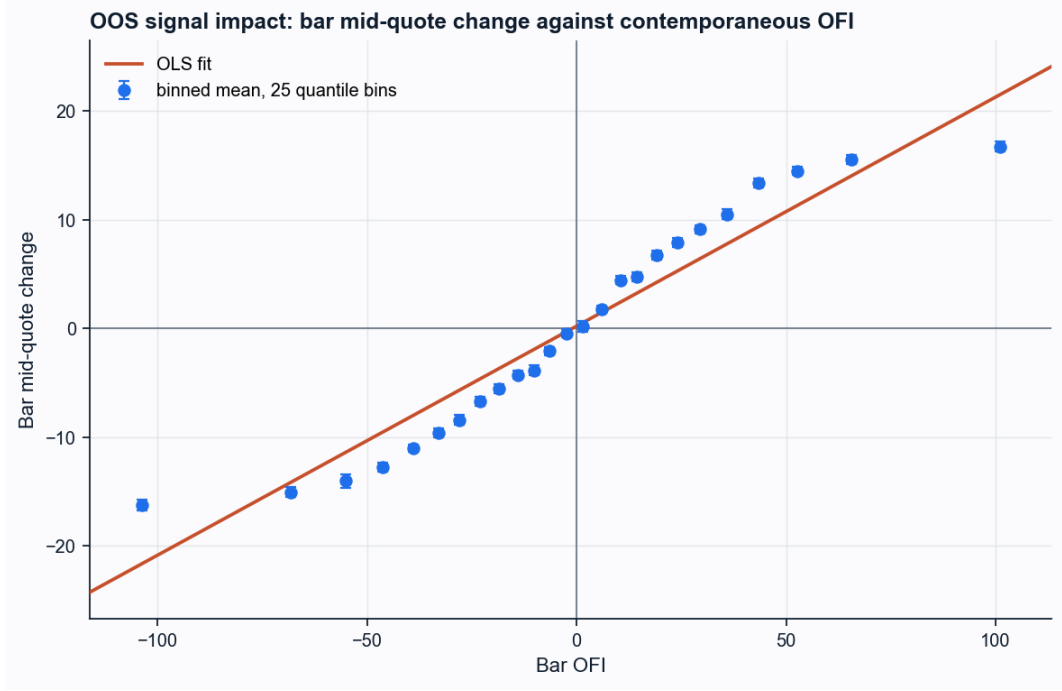


Figure 1: Out-of-sample bar mid-quote change against contemporaneous OFI, 25 quantile bins with ± 1 standard error and the OLS fit. The relationship is close to linear; full estimates are in Table 1.

first reaches $|z_t| \geq 1$, with μ_t and σ_t the trailing mean and standard deviation of C over a 200-bar window shifted by one bar. The sign of z_t gives the event direction. This defines a timestamp and a direction only; no position, holding rule, or cost is involved.

For each of the 5,099 OOS signal events, the mid-quote at $t = 0$ and at each horizon is taken from the raw TBBO, and the cumulative return is signed by event direction. A bar's `mid_close` is the mid at the last event inside its final minute. Measurements where the matched BBO at the target time is more than 120 seconds stale (e.g. past the daily 5–6 pm ET session break) are dropped, so the sample falls with horizon.

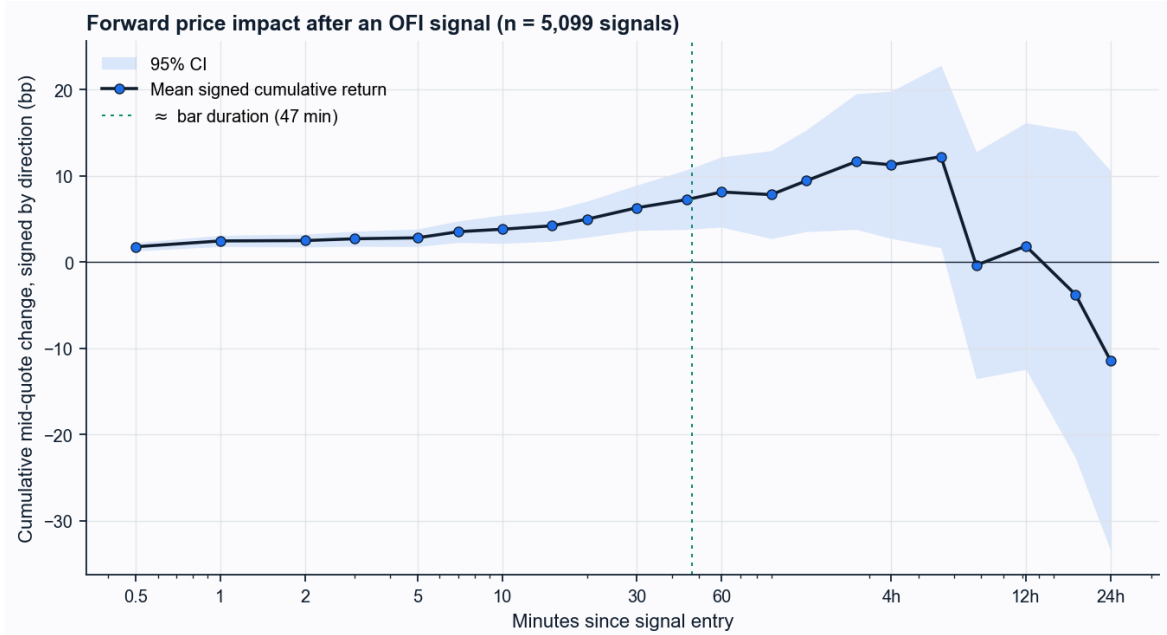


Figure 2: Signed forward return after a signal event, OOS. Peak +11.3 bp at four hours; not distinguishable from zero beyond it.

Horizon	Signed return (bp)	Block-bootstrap t	95% CI (bp)	n
0.5 min	+1.78	7.1	[+1.3, +2.3]	5,099
1 min	+2.45	7.5	[+1.8, +3.1]	5,099
2 min	+2.50	6.6	[+1.7, +3.3]	5,099
10 min	+3.81	4.6	[+2.2, +5.5]	4,851
30 min	+6.30	4.7	[+3.8, +8.9]	4,758
1 hour	+8.13	3.9	[+4.0, +12.2]	4,644
2 hours	+9.44	3.1	[+3.5, +15.2]	4,236
4 hours	+11.28	2.6	[+2.9, +20.1]	4,159
8 hours	-0.36	-0.1	[-13.1, +12.9]	3,519
12 hours	+1.86	0.3	[-12.1, +16.6]	3,619
24 hours	-11.46	-1.0	[-32.6, +11.5]	3,293

Table 2: Mean forward signed return at fixed horizons after an OOS signal event. The t -statistic and 95% interval are from a by-day block bootstrap (2,000 resamples), which keeps each day’s overlapping forward windows together. A non-overlapping subsample (events at least one horizon apart) gives $t = 2.4$ at the four-hour peak.

Figure 2 plots the profile and Table 2 reports it. A sign-randomised placebo at the same timestamps, averaged over 50 seeds, stays within ± 1.3 bp of zero at every horizon and its 95% band always covers zero; the OFI-direction curve is not market drift. Standard errors and confidence intervals are from a block bootstrap over trading days (2,000 resamples): the forward windows of nearby events overlap, so a cross-sectional standard error understates uncertainty at long horizons. A non-overlapping subsample, keeping events at least one horizon apart, is reported as a cross-check.

The peak remains significant after the overlap correction (block-bootstrap $t = 2.6$, 95% CI [+2.9, +20.1] bp; non-overlapping subsample $t = 2.4$); the cross-sectional statistic overstates it, and beyond four hours the signed return is not distinguishable from zero.

4 Economic magnitude

The decay profile is statistically clear but economically small relative to the cost of transacting on this contract. The four-hour peak is the largest signed move the signal anticipates; by ten minutes the figure is +3.8 bp. Against this, the front-month quoted spread on CME Ether futures runs 10 to 30 ticks over the sample, or roughly 1.7 to 5.0 bp per side at typical prices, so a single round trip that crosses the spread consumes a substantial fraction of the entire four-hour move and more than the whole ten-minute move.

5 Conclusion

The OFI signal is present on CME Ether futures at HAC $t = 58.1$ OOS. The main empirical contribution is the decay profile, significant at every horizon out to the four-hour peak, flat under a sign-randomised placebo, and indistinguishable from zero beyond the peak. The predictable move is comparable to the quoted spread, so the relationship is informative about price formation without being straightforwardly tradable.

The construction is level-1 only: the TBBO source carries the touch and trades but not deeper book levels, so the deeper-book dynamics that the original equity result relies on cannot be measured here. TBBO is also trade-sampled, meaning the book is observed only when a trade prints. A quote-level feed would sharpen the sub-minute horizons and permit a proper treatment of execution, which is the natural extension.

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